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Who Goes Next? The Gendered Expansion of Mexican and Senegalese Migrant Sibling Networks in Space and Time

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The migration literature shows that individuals whose siblings have migrated abroad are more likely to migrate, yet we know little about sibling migrant networks. We use MMP and MAFE-Senegal survey data to compare migration patterns in two very disparate contexts (Mexico and Senegal) in an attempt to assess the scope, manner, and generalizability of sibling network migration patterns. Our results show that while Senegalese families are likely to have one international migrant, Mexican families are likely to send two or more members abroad. Sibling migrations from Mexico fall closer together in time than do those from Senegal, suggesting joint sibling migration. Also, while Mexican sibling networks did not seem to contribute to the expansion of Mexican migrant destinations, Senegalese sibling networks did contribute (slightly) to the expansion of Senegalese migration. Sibling networks in both settings contributed considerably to the feminization of migration.

Keywords: migration; family; gender; siblings; geography; Mexico; Senegal

One of the most consistent findings of empirical migration research around the globe is that people are considerably more likely to migrate internationally when they are connected to other individuals who have moved abroad before: these people include fellow community members, acquaintances, friends, or relatives (Boyd 1989; Liu 2013; Massey and Aysa-Lastra 2011; Massey and Riosmena 2010).

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Prior migrants can provide general as well as destination- or labor market-specific social capital to prospective migrants (Portes 1998) and are likely sources of potential or actual resources—information, job connections, or other forms of settlement assistance—that facilitate the migration of new arrivals to settlement and destination areas (Flores-Yeffal 2013; Menjívar 2000; Sue et al. 2018).

People in sending communities often have access to such social capital since migrants-to-be and previous migrants are normatively bound to help each other via reciprocity or mutually enforceable trust rules stemming from their membership and position in more or less durable networks of institutionalized relationships (Portes 1998). Of these, blood connections are exceptionally important, with sibling ties being perhaps the most valuable. Research shows that brothers and sisters stand out as being particularly effective in facilitating the subsequent migrations of their siblings (Palloni et al. 2001; Massey and Espinosa 1997; Massey and Aysa-Lastra 2011; Liu 2013).

Individuals may be more likely to draw on social capital emanating from sibling ties since strong normative expectations generally require that brothers and sisters help one another in times of need; this norm continues to exist in most, if not all, societies, especially in low- and middle-income nations (e.g., Cicirelli 1994). This process is often highly gendered (Curran and Saguy 2001), with brothers being most likely to receive help from a migrant sibling of either sex, and brothers being most likely to sponsor one other (Curran and Rivero-Fuentes 2003; Liu, Riosmena, and Creighton 2018). As such, sibling networks are fundamental conduits by which broader migrant networks expand over both time and space.[1]

The sequential migration of siblings—where siblings subsequently migrate to the same destination with help from siblings who previously migrated—is clearly in line with social capital theory (Portes 1998; Massey et al. 1993). However, it may or may not be consistent with other theoretical models such as the new economics of labor migration (NELM), in which migration is seen as a strategy used by families to diversify their exposure to risk (e.g., that stemming from economic uncertainty, political upheaval, or environmental change) by distributing their labor resources across geographically distinct markets (Massey et al. 1993; Taylor 1986; Sana and Massey 2005; Stark and Bloom 1985). Although many different patterns of temporal spacing and geographic sequencing in the migration of siblings are consistent with coordinated decision-making on the part of family members, the diversification of risk requires different siblings to migrate either to different locations at one point in time or to the same location at different points in time to be effective (Borjas and Bronars 1991; Palloni et al. 2001), underscoring the importance of timing and sequencing underlying motivations for migration.[2]

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Despite the clear relevance of the pace and sequencing of network expansion to the identification of different family migration strategies, we actually know little about the temporal, geographic, and gendered distribution of migrants within households, or whether these indeed support risk-diversification expectations. Here, we use data from two geographically and culturally distinct contexts—Mexico and Senegal—to examine how the timing and sequencing of migration by siblings are differentially structured with respect to kinship, gender, and birth order, which are important axes of social action in both settings (see Bass and Sow 2006; Gabrielli 2010; McGinn 1966; Liu et al. 2016). We hope that a systematic comparison of migration patterns in two very disparate contexts will help to assess the scope, manner, and generalizability of sibling network migration patterns.

Sibling Networks, Social Capital, and Migration

Migration by one person in a social network influences the likelihood that others in the same network will also move by providing information and resources that facilitate migration, with sibling networks being particularly effective conduits (Garip 2008; Liu 2013). Prior studies of sibling migrant networks provide a foundational understanding of how migrant networks are gendered in different cultural contexts. In Mexico, although sister-initiated migration is uncommon and brothers tend to sponsor each other's migration, the prior migration of brothers does lead to sister migration as well (Curran and Rivero-Fuentes 2003). In Senegal, however, the prior migration of a sibling appears to benefit a sister's and a brother's migration more equally (Liu, Riosmena, and Creighton 2018), whether authorized or unauthorized (Liu, under review).

Despite being an important (if generally implicit) element in many studies of international migration, comparatively little work has been done on the pacing and sequencing of migrant siblings within households. Instead, studies have focused mainly on assessing the predictive power of sibling networks (Borjas and Bronars 1991; Palloni et al. 2001; Liu et al. 2018; Massey and Riosmena 2010), but less so in the manner in which the expansion of these networks has taken place. For example, in their study of Mexican brother pairs, Palloni and colleagues (2001) find very important sibling network effects for younger brothers, for whom having an older brother migrate nearly tripled their likelihood of migration (in contrast, older brothers did not seem to benefit from a younger brother's migration at all, as they exhibited similar migration rates to those of someone lacking networks entirely). Likewise, in their study of Mexican and Senegalese sibships, Liu and colleagues (2016) find strong migrant sibling network effects in both settings. We expand on previous work by examining how sibling networks distribute across time and space.

The way in which migrant networks are distributed across different geographies and years of arrival is expected to affect the quality and amount of information and resources they can provide. Prospective migrants may anticipate in-kind resources (housing, food, transportation) from network members currently

residing in the specific destination areas they target (Flores-Yeffal 2013). Indeed, new migrants often choose destinations based on the presence of ties and, especially, in the expectation that some social ties will be more valuable than others in providing assistance during a moment of need (Sue et al. 2018). Likewise, information about navigating migration routes, gaining access to work, and the logistics of daily life are typically place- and time-specific. Few studies have examined the structure of migrant networks in terms of destination (Curran and Rivero-Fuentes 2003; Toma and Vause 2014) or their timing.

Comparative Perspectives on Migration from Mexico and Senegal

Mexican and Senegalese family systems and migration dynamics share broad similarities that suggest sibling migrant networks operate in similar ways. Siblinghood strongly structures social obligations in both settings (for Mexico, see McGinn 1966; for Senegal see Bass and Sow 2006; Gabrielli 2010) and migration by, then assisted by (often, older), siblings may be used as a vehicle to fulfill those obligations (see Liu et al. 2016). However, Mexico and Senegal are strikingly different in terms of kinship and gender roles and, hence, the family systems they generate and the patterns of international migration that result (Garip 2016; Massey, Durand, and Malone 2002; Schoumaker et al. 2013; Liu 2013). These differences may not only produce different migration intensities for families across the two contexts, but also generate dissimilar network structures with respect to timing, sequencing, gender, and geography, especially among siblings.

In terms of family structure, polygyny is relatively common and socially accepted in Senegal but not in Mexico, where it is not legally or socially sanctioned and only occurs surreptitiously and less commonly. As Senegalese offspring are often distributed across various households (and economies), polygny complicates assumptions about siblings occupying a common parental household and decision-making unit. It also makes relations among siblings more complex, and possibly more diffuse, perhaps leading to a lower influence of sibling networks in Senegal—particularly in coordinating the timing and destination of moves—relative to the Mexican case.

Regarding gendered family systems, while most ethnic groups in Senegal follow patrilineal traditions, some exhibit matrilineal patterns. This contrasts, at least slightly, with Mexico, where family systems are more consistently patriarchal and patrilineal. If gendered migration and migrant network influences vary among patrilineal and matrilineal family systems in Senegal (Liu and Lindstrom, under review), sibships may be less dominated by brothers in Senegal than in the Mexican case.

Mexico and Senegal also differ in terms of the intensity and geography of their international migrations. Mexican migration has historically involved larger proportions of the population, with about one-tenth of the Mexican-born population living

abroad (Fitzgerald 2008). In contrast, Senegalese migration is lower in intensity, with only 4 percent of all Senegalese-born individuals living abroad (Ratha and Xu 2007); though one study reports that around one in five Senegalese have lived abroad during their lifetime (Schoumaker et al. 2013). Here, we assess the ways in which differences in international migration intensity relate to the degree and manner in which sibling migrations are clustered. More specifically, we examine whether higher migration intensities in Mexico are due to a higher likelihood that a family will "send" any sibling abroad, or due to a higher likelihood that a family that has previously sent one or more siblings abroad does so again, while also examining differences in the timing and gender and spatial patterns between countries.

Senegalese and Mexican migration differ considerably in terms of their scale of spatial concentration. An overwhelming majority of Mexican migrants (98 percent) live in the United States (World Bank 2010), though since the 1990s these destinations have diversified considerably, with the share of migrants going to traditional gateways in California, Texas, and Illinois (the Chicago Metropolitan Area) decreasing somewhat, owing to the reemergence and rise of other destinations spread out across the country (Durand, Massey, and Capoferro 2005).

In contrast, national destinations for Senegalese migrants have historically been much more geographically diverse, with migrants going to several African and European nations by the mid- to late twentieth century, and with smaller, nontrivial, portions going increasingly to other countries as well, including the United States. Although Senegalese migration has also expanded geographically in recent decades, this dispersion, unlike the Mexican case, has transcended national borders, moving more into, and within, Europe. While African destinations dominated Senegalese out-migration immediately following Senegal's 1960 independence from France, the share of migrations to Europe grew from 40 to 60 percent between 1974 and 2008 (Schoumaker et al. 2013). In this study, we assess whether the expansion of migrant sibling networks has contributed to the diversification of destinations for Mexican and Senegalese migrants, or whether divergence away from traditional area destinations occurred despite, rather than because of, sibling network expansion.

Data and Methods

We employ two roughly comparable sources of household survey data about migration: the Mexican Migration Project (MMP) and the Migration between Africa and Europe–Senegal (MAFE) project (see Liu et al. 2016 for more details on their comparability). Since 1982, the MMP has surveyed four to six Mexican communities per year, drawing a simple random sample of the whole community in smaller rural localities and towns, the core area of larger towns, or one or several neighborhoods in larger cities. In our analyses, we use eighty-three communities surveyed between 2000 and 2015, restricting the temporal span of data collection to keep it relatively close to and centered on that of the MAFE samples, which were collected in 2008.

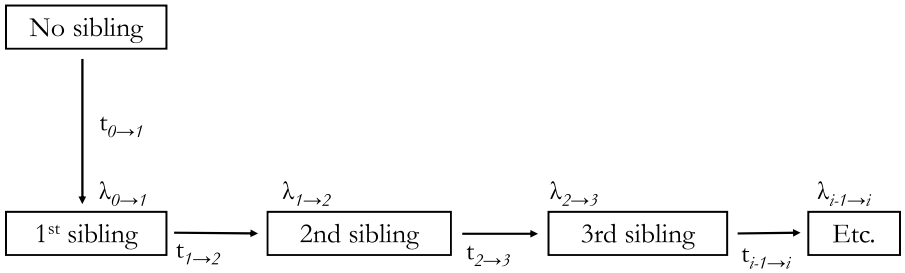
During this year, MAFE-Senegal drew a two-stage stratified random sample of households in greater Dakar (Beauchemin 2012; Willekens, Zinn, and Leuchter 2017). The MAFE–Senegal household data are representative of the greater Dakar region, which is home to about a quarter of the country's population, including most of its urban dwellers, and serves as the source of 31 percent of international migrants reported by Senegalese households in the 2001–2002 nationally representative ESAM II survey (Sall 2008).

MMP and MAFE respondents were surveyed using flexible household questionnaires (known as ethnosurveys), which compiled socioeconomic, demographic, and retrospective migration information about the household head, the head's spouse, and—importantly for this study's purpose—*all* children of the head regardless of their household status at the time of survey (thereby allowing us to measure the behavior for all siblings). Retrospective migration information on siblings included year of first migration, destination (reflecting the geography of Mexican and Senegalese migration, place in the United States in MMP, and country of destination in MAFE), year of first return, and place of current residence. Given our interest in sibling networks, we draw from an analytical sample of 795 Senegalese and 7181 Mexican sibships, comprising a total of 3,705 Senegalese and 28,423 Mexican siblings.

Our sample includes children of both male- and female-headed households. About 15 percent of Mexican households and 36 percent of Senegalese households in our samples were headed by women. We have a full roster of all children of the household head (whether they are coresident or not) at the time of survey. Full lists of other siblings (e.g., other children of any parent who was not the "head" of the surveyed household, children of a nonresident parent, or children of a resident step-parent) are not available. While the resulting data suffice for examining risk-diversification explanations that usually rely on definitions of household roles such as head, spouse, and children of head, other uncounted siblings may be sources of valuable information and resources of migration. If so, our study and prior analyses relying on family or sibling data collected in a similar way may underestimate the actual influences of sibling networks.

Our analysis follows two main approaches. First, we analyze the intensity, pacing, and sequencing of migration within sibships using descriptive techniques akin to the parity progression ratios employed in fertility research (see Henry 1953/1980). These ratios indicate 1) the probability that a family "sends" at least $i+1$ siblings abroad; and 2) the probability that a family sends a $i+1^{\text{th}}$ sibling abroad given that it had previously sent i siblings abroad (see Figure 1 for a visual depiction of this state-space). In addition to these intensities, we examine the characteristics of siblings according to their migration order, including their composition in terms of sex, age at migration, year of migration, destination (for Senegal, Europe vs. Africa vs. other; for Mexico, traditional vs. newer destinations).[3] We also compare these characteristics between i^{th} and $i+1^{\text{th}}$ migrant siblings and, in addition, measure the duration of time between siblings' migrations as well as the gender and destination distribution for the $i+1^{\text{th}}$ sibling, given the particular gender and destination type for sibling i.

FIGURE 1
State-Space for the Study of Migration Progression and Interval between the Migration of Siblings



Assessing the timing and spatial correspondence between sibling migrations helps us to better distinguish whether and when sibling networks are being used as more direct sources of assistance (e.g., migration to similar places within a relatively short interval), or how much sibling migration may be used to distribute risk (e.g., among different destinations, or across longer time periods).

Measuring the Timing and Sequencing of Sibling Migration

Table 1 shows descriptive statistics pertaining to migration "parity" among Mexican and Senegalese sibships. Senegalese families are slightly more likely than Mexican families to send one sibling abroad, with 32 percent of the former and 27 percent of the latter doing so (see Figure 2 for a visual depiction). In contrast, as described before, Mexican families are more likely to send subsequent siblings abroad, partially explaining the higher rates of migration in Mexico relative to Senegal. Roughly half of Mexican families and around 40 percent of Senegalese families that had i siblings with prior migration experience, saw an $i+1^{th}$ sibling migrating internationally with or after them (see Figure 3).

While these figures show the power of migrant sibling networks, they also suggest that only a minority of sibships have multiple migrants: in the case of Mexico, 14 percent and 7.6 percent of sibships had two or more and three or more U.S. migrants, respectively, with figures for Senegalese sibships being slightly and somewhat lower at 12.5 percent and 5.4 percent, respectively. Indeed, only a small minority of families engage in behavior that—depending on the timing, geography, legal status, and other characteristics not directly observed in our data—may fit with so-called chain migration patterns (although also note that many of these individuals returned to the sending country before the survey year).

To better assess the type of sequential migration $i+1^{th}$ migrant siblings are engaging in, Table 2 (divided into panels A and B, for Mexico and Senegalese families, respectively) shows more specific results on the pacing and sequencing of sibling migration. The average age at migration is about 22 years old for the

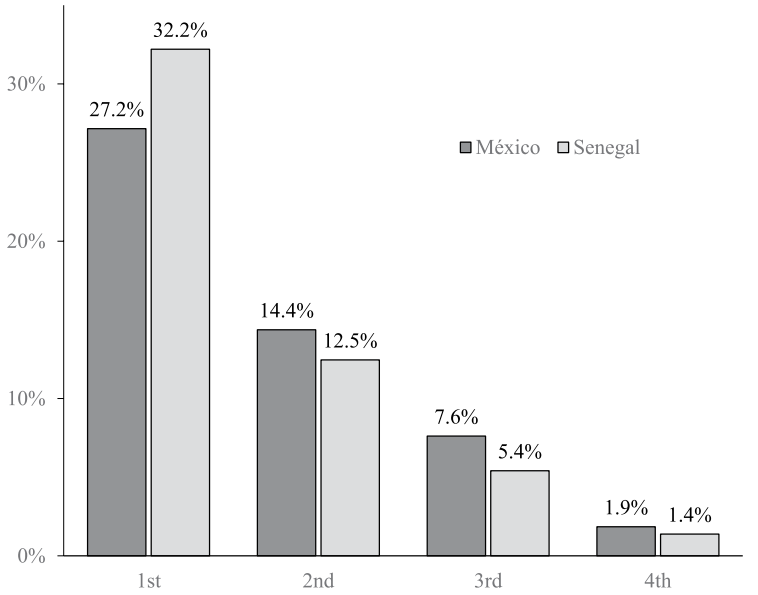
TABLE 1

Descriptive Statistics of Migration "Parity" among Mexicans and Senegalese Siblings (MMP and MAFE data)

Mexican Migration 'Parity'	0	→	1	→	2	→	3	→	4
No. of sibships with *i* migrants	5231		918		485		281		133
No. of sibships with at least *i* migrants	7181		1950		1032		547		266
					Proportions (and standard errors)				
Prop. of sibships with exactly *i* migrants	0.728 (0.006)		0.128 (0.011)		0.068 (0.011)		0.039 (0.012)		0.019 (0.012)
Progression (0, *i*)			0.272 (0.015)		0.144 (0.016)		0.076 (0.016)		0.037 (0.016)
Progression (*i*-1,*i*)			0.272 (0.015)		0.529 (0.023)		0.530 (0.030)		0.486 (0.043)
Senegalese Migration 'Parity'	0	→	1	→	2	→	3	→	4 →
No. of sibships with *i* migrants	539		157		56		19		13
No. of sibships with at least *i* migrants	795		256		99		43		24
					Proportions (and standard errors)				
Prop. of sibships with exactly *i* migrants	0.678 (0.020)		0.197 (0.032)		0.070 (0.034)		0.024 (0.035)		0.016 (0.035)
Progression (0, *i*)			0.322 (0.037)		0.125 (0.044)		0.054 (0.052)		0.030 (0.047)
Progression (*i*-1,*i*)			0.322 (0.037)		0.387 (0.065)		0.434 (0.114)		0.558 (0.138)

SOURCE: Mexican Migration Project and Migration between Africa and Europe Project.

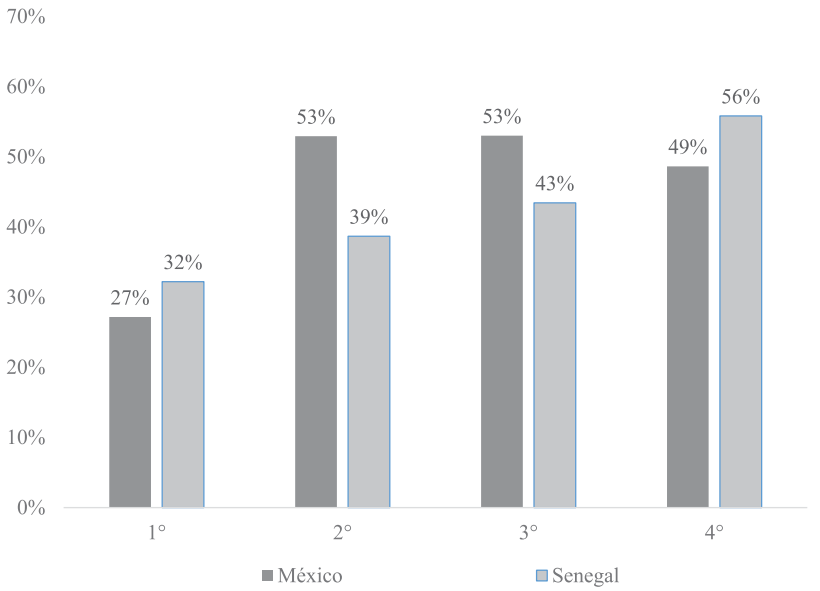
FIGURE 2
Likelihood that Sibling Becomes the "i"th Migrant of Sibship



first migrant sibling in Mexico, increasing only very slightly for second and third siblings (22.3–22.4), and further increasing a bit for fourth migrant siblings (23.0, see Table 2A). First international migrations occur slightly later among Senegalese siblings than Mexican ones (e.g., 22.6 years of age for the first migrant sibling), and they do not increase monotonically by sibling migration order, fluctuating between 22.0 and 24.4 years across the different migrant sibling orders. In general, this means that the social capital potentially available to individuals via their sibling networks does not appear to accelerate the timing at which Mexicans and Senegalese initiate their migratory "careers." While MAFE did not collect data on the marital status of each sibling at the time of first migration, note that these ages at first migration are generally lower than the mean age at marriage in both countries, particularly for men.[4]

In term of pacing, Table 2 also shows that sibling migrations appear to fall closer together in time in Mexico than in Senegal, especially for the second migrant sibling, who, on average, migrated almost four years after the first migrant in the sibship in the Mexican case, with the corresponding figure being almost seven years in Senegal. In contrast, people with two or three siblings migrating previously to them only waited 3.1 and 2.9 years in Mexican families, and 2.4 and 3.2 years in Senegalese sibships. To further illustrate, Figure 4 shows a more explicit distribution of time intervals between (A) first and second migrant siblings and (B) second and third migrant siblings, respectively, showing Mexican families on the left side and Senegalese on the right side of the figure. Indeed, in Mexico, around a fifth of families "sent" two siblings the same year, either as the

FIGURE 3
Conditional Probability that $i+1^{th}$ Sibling in Family Migrates, Given that i Sibling(s) Had Migrated Previously



pioneers in the sibship (Figure 4A) or after a first pioneer sibling had migrated (Figure 4B). While these percentages are similar for the transition between second and third migrant within Senegalese sibships (Figure 4B), note that the share of Senegalese sibships sending two or more siblings abroad in the same year is somewhat lower in families that had not sent a migrant sibling prior to those migrations, at 8 percent (Figure 4A). Indeed, of the Senegalese families who had sent at least one international migrant sibling, the migrations of a second sibling occurred 10 or more years after the first one for 29 percent, an amount 2.6 times larger than the 11 percent of Mexican sibships, with this decade-or-longer gap.

The spatial sequencing of sibling migration

While these figures suggest that a nontrivial number of Mexican siblings migrated in tandem (including not only those siblings migrating the same year, but also the 15–20% of migrant siblings moving the year after the prior sibling had done so), to infer some of the possible motivations behind them, it is important to examine whether these migrations were also spatially synchronized. As shown in both Table 2 and Figure 5, the amount of spatial match between siblings was fairly high in Mexico in general and for higher-order Senegalese migrant siblings.

In the Mexican case, around 75 percent of second, third, and fourth migrants in the sibship went to the same U.S. state as the prior migrant sibling;

TABLE 2
Pacing and Sequencing of Mexican and Senegalese Sibling Migrations

	1st migrant		2nd migrant		3rd migrant		4th migrant	
A. Mexico (MMP)	Mean	(S.E.)	Mean	(S.E.)	Mean	(S.E.)	Mean	(S.E.)
Age at first international migration	22.0	(0.063)	22.3	(0.082)	22.4	(0.104)	23.0	(0.145)
Time between prior and current's sibling's first migrations (see Figures 4 and 5 for distributions)			3.9	(0.053)	3.1	(0.060)	2.9	(0.068)
Male (REF = female)	0.755	(0.004)	0.718	(0.005)	0.671	(0.007)	0.643	(0.009)
Current mig. sibling male | prior migrant sibling male			0.767	(0.005)	0.698	(0.008)	0.684	(0.010)
Current mig. sibling female | prior mig. sibling male			0.140	(0.003)	0.183	(0.005)	0.179	(0.006)
Current mig. sibling male | prior mig. sibling female			0.582	(0.010)	0.605	(0.012)	0.565	(0.015)
Current mig. sibling female | prior migr. sibling fem.			0.418	(0.010)	0.395	(0.012)	0.435	(0.015)
Destination Composition: Traditional	0.675	(0.004)	0.697	(0.005)	0.716	(0.006)	0.742	(0.008)
Newer	0.325	(0.004)	0.303	(0.005)	0.284	(0.006)	0.258	(0.008)
Migrated to same U.S. state as prior migrant sibling			0.772	(0.005)	0.774	(0.006)	0.775	(0.008)

	1st migrant		2nd migrant		3rd migrant		4th migrant	
B. Senegal (MAFE).	Mean	(S.E.)	Mean	(S.E.)	Mean	(S.E.)	Mean	(S.E.)
Age at migration	22.6	(0.234)	22.0	(0.355)	24.4	(0.558)	23.0	(0.627)
Time between prior & current's sibling's first emigrations (see Figures 4 and 5 for distribution).			6.9	(0.277)	2.4	(0.161)	3.2	(0.249)
Male (REF = female)	0.662	(0.012)	0.555	(0.019)	0.479	(0.028)	0.570	(0.036)
Current mig. sibling male | prior mig. sibling male			0.593	(0.025)	0.481	(0.040)	0.481	(0.057)
Current mig. sibling female | prior mig. sibling male			0.145	(0.011)	0.223	(0.022)	0.256	(0.035)

TABLE 2 (CONTINUED)

	1st migrant		2nd migrant		3rd migrant		4th migrant	
B. Senegal (MAFE).	Mean	(S.E.)	Mean	(S.E.)	Mean	(S.E.)	Mean	(S.E.)
Current mig. sibling male | prior mig. sibling female			0.500	(0.031)	0.476	(0.039)	0.633	(0.046)
Current mig. sibling female | prior mig. sibling female			0.500	(0.031)	0.524	(0.039)	0.367	(0.046)
Destination Composition: Europe	0.552	(0.012)	0.610	(0.019)	0.637	(0.027)	0.624	(0.036)
Africa	0.343	(0.012)	0.287	(0.018)	0.211	(0.023)	0.274	(0.033)
Other	0.105	(0.008)	0.103	(0.012)	0.152	(0.020)	0.102	(0.022)
Migrated to same region as prior migrant sibling			0.576	(0.008)	0.831	(0.006)	0.917	(0.004)

SOURCE: Mexican Migration Project and Migration between Africa and Europe Project.

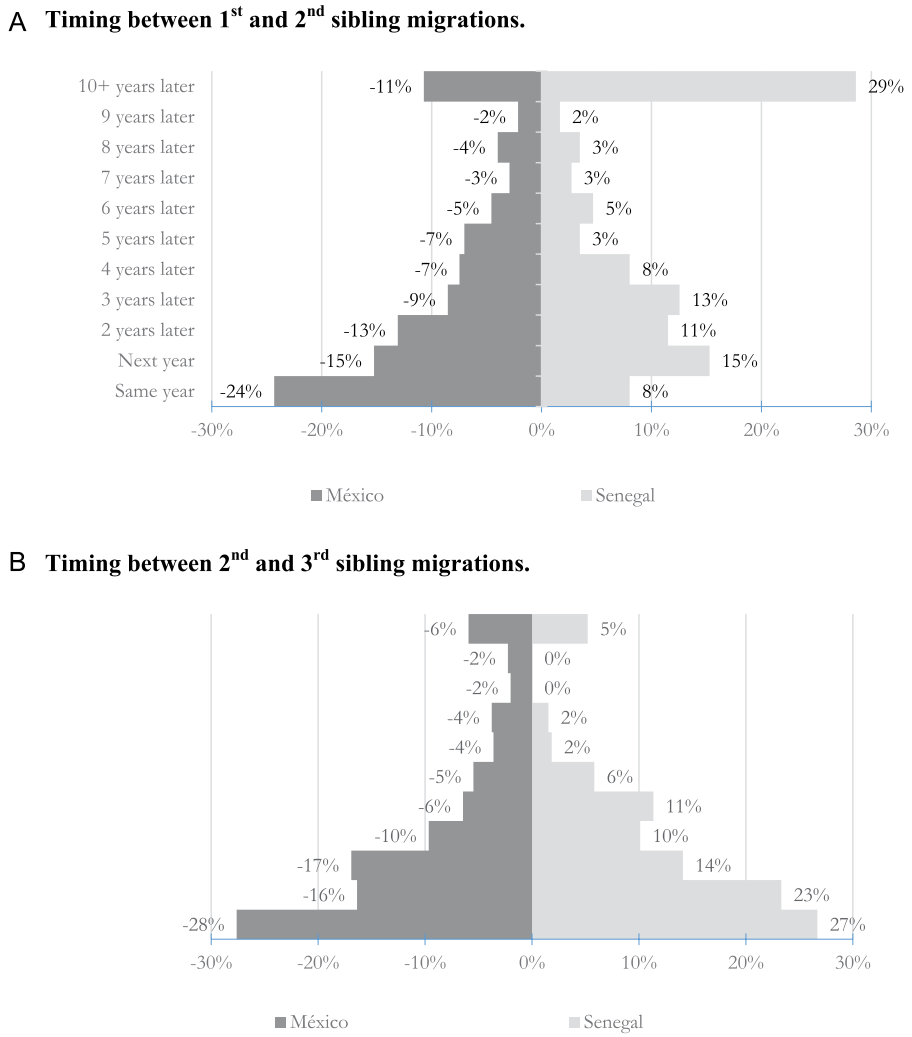
also note that the percent of migrant siblings going to traditional (as opposed to new/re-emerging) U.S. destinations was a bit lower for the pioneer migrants in the sibship than for all other migrant siblings. That is, over two-thirds of migrant pioneer siblings went to a traditional U.S. destination, while the percentage was a little higher for second, third, and fourth migrant siblings at 71–73 percent. This suggests sibling networks were either *not* instrumental in the expansion of Mexican migration to new U.S. destinations throughout the 1990s and early 2000s in particular (Zúñiga and Hernandez-León 2005), or that the composition of multimigrant sibships is tilted toward pre-expansion times.

In contrast, in Senegal, "only" 58 percent of second migrant siblings went to the same *world region* as the pioneer migrant in the sibship (though the percentage of third and fourth migrants in the sibship going to the same region and second and third ones was much higher, at 83 percent and 91 percent, respectively). As such, it is quite likely that the mismatch between the destinations of first and second migrant siblings in Senegal is, at least partly, a cohort "effect" likely due to a relative increase in migration to Europe relative to Africa after 1990 (see Schoumaker et al. 2013).

Gendered patterns of migration within sibships

Reflecting the overall patterns of migration, sibling migrations from both Mexico and Senegal are heavily gendered. While brothers dominate sibling migrations from both countries, male-dominated patterns are particularly accentuated in the Mexican case for both pioneer and subsequent

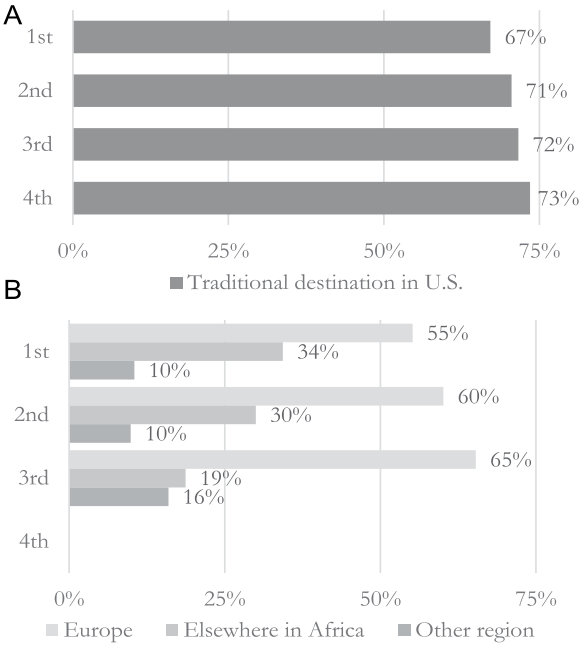
FIGURE 4
Distribution of Time Intervals between Migrations for Mexico and Senegal



(A) 1st and 2nd Siblings
(B) 2nd and 3rd Siblings

migrant siblings. As shown in Figure 6, about 75 percent of first Mexican migrant siblings are male, with the proportion of males decreasing with migration order: 72 percent of second sibling migrants are male, as are 68 percent of third sibling migrants, and 63 percent of fourth sibling migrants. In Senegal, sibling migration is still male-dominated among lower-order migrants, but less so than the Mexican case: about 66 percent and 56 percent of first migrant

FIGURE 5
Geographic Distributions of Sibling Migrations by Sibling Migration Order

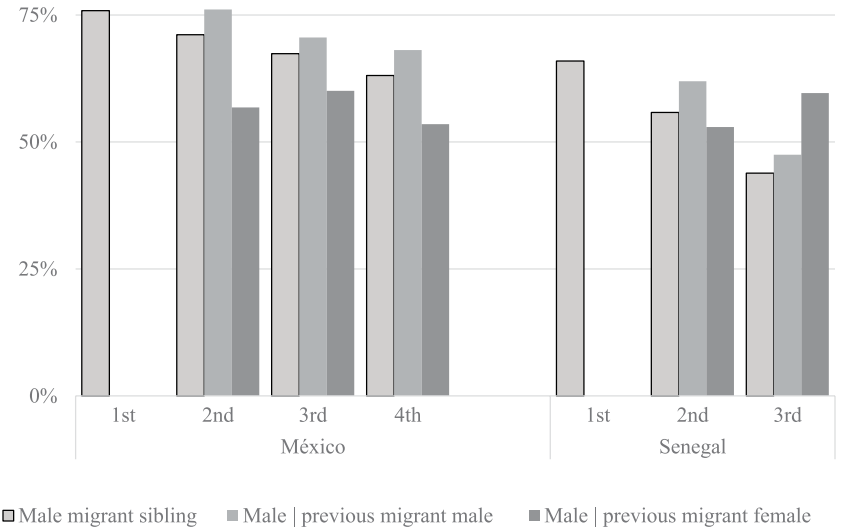


(A) Mexico
(B) Senegal

siblings are male. However, note that only 44 percent of third siblings are male, indicating a stronger "feminization" of sibling networks in the Senegalese than the Mexican case.

Despite this feminization, the broader patterns do suggest that within-sibship dynamics help to reproduce male-dominated migrant networks in both settings (but again, especially in Mexico). Also shown in Figure 6, in about 76 percent of Mexican sibships where the pioneer migrant was a brother, the second migrant in the sibship was also a brother. Even in the smaller share of Mexican sibships where the pioneer migrant was a sister, a majority of the second migrants were brothers at about 58 percent. In Senegal, the brother-dominated pattern exists but is indeed not as marked: in 62 percent (53 percent) of sibships where the pioneer migrant was a brother (sister), the second migrant was also a brother. As established by prior studies on both settings (e.g., Liu et al. 2016), brother migration begets further brother migration. While sister migration is also usually followed by brother migration, having a sister migrate first does boost the probability that other sisters will migrate.

FIGURE 6
Gender and Migration Likelihood, by Sibling Migration Order for Mexico and Senegal



Conclusion

This cross-context comparative study shows the manner in which migrant sibling networks expand over time and space in gendered ways. We find that Senegalese families are more likely to "send" a pioneer migrant than Mexican families, and that Mexican families are more likely to have subsequent siblings migrate than the Senegalese. This explains part of the higher propensity of Mexicans to migrate internationally relative to Senegalese. In this sense, Mexican sibling migrant networks appear to be more effective facilitators of mobility than Senegalese ones, perhaps because of the longer history of Mexican migration and because Mexican out-migration has reliably been concentrated in one single country of destination. In addition to this possibility, an examination of the pacing between migrations also suggests that Mexicans are more likely to migrate along with other siblings than Senegalese migrants, at least for the first two migrants in the sibship.[5] This tighter timing of sibling migrations may signify more effective sibling migrant networks in the Mexican case, and also important differences in the ways individuals and families make migration decisions in Senegal and Mexico.

Indeed, our study contributes to the understanding of how family members engage in migration potentially as a collective strategy, which is an important tenet in migration studies, especially under the new economics of labor migration theory. While scholars generally argue that the household risk aversion goals predicted by NELM is attained through risk diversification, specifically moves that are spaced out in time and/or geography, our findings here suggest otherwise, particularly in Mexico. Given that labor migrations are often predicated

upon a clear promise of work opportunities by network members (e.g., Sue et al. 2018), our study suggests that the migration of siblings to similar places at the same or about the same time represents a different form of risk aversive behavior that is not consistent with location-related or time-graded diversification. While siblings migrating together or almost together can be participating in different labor markets (in terms of, for example, occupations), previous research suggests that earlier arrivals who help more recent migrants connect to jobs do so in occupations that are similar to those in which they work (Flores-Yeffal 2013; Krissman 2000). As such, some families may indeed coordinate migrations in ways that do not necessarily lead to risk diversification but do indeed lead to some forms of risk aversion.

Note that, while our findings question the general notion of risk diversification, they do not disprove many other aspects of NELM, as sibling migrations could indeed still be the result of joint family strategies to avert risk brought by imperfect markets in sending areas. Likewise, our study was not designed to test whether relative deprivation—an important NELM mechanism—motivated people to move, though we note that people may feel relatively deprived by the improvement in economic conditions not only of other migrants in the sending community, but of their siblings in particular. Future research—especially qualitative work—should continue to address these questions more explicitly and investigate the degree to which family members coordinate moves and the motivations behind these joint decisions.

Our study also contributes to the understanding of gender and migration by showing that sibling migration sequencing is highly gendered and masculinized, even when the prior migrant in the sibship is a sister. In our study, this was particularly true in Mexico. However, we also note that the "feminization" of migration does occur as more siblings migrate—a finding especially true among the Senegalese. As such, sibling networks are likely an instrumental conduit by which labor migration diversifies in terms of gender, in addition to "tied migrations" due to family reunification (Cerrutti and Massey 2001; Riosmena 2010; Toma and Vause 2014).

In contrast with our findings about gender, sibling networks do *not* appear to have been important contributors to the geographic expansion of Senegalese and, especially, Mexican migration. In the Mexican case, subsequent migrant siblings were as likely, or even more likely, to migrate to traditional Mexican immigrant gateways than the migrant pioneer in the sibship. In Senegal, the second migrant sibling was considerably less likely to go to the same world region as the sibling pioneer, who, often in the data, had migrated several years before. This is perhaps a result of a structural shift where increasingly large numbers of migrants (and subsequent sibling migrants among them) headed to Europe than in the past, when a larger share of these flows were directed to other African nations. In addition to these shifts or cohort effects in the Senegalese case, the relative absence of siblings as instrumental conduits for destination expansion could suggest that weaker ties may have been particularly instrumental in this process, as siblings may operate as strong ties most often due to normative expectations of mutual solidarity and reciprocity that govern the sibling relation. Future research

should continue to understand the type of individuals and families that migrate in tandem (in more or less gendered ways), and their spatial and occupational adaptation.

Migrant networks—especially family connections to migrants—are instrumental components of the migration process. A large body of work establishes this relevance both quantitatively and qualitatively in terms of their role in increasing the likelihood of migration, destination choice, and settlement outcomes (Flores-Yeffal 2013; Garip 2016; Liu 2013; Massey and Aysa-Lastra 2011; Massey and Riosmena 2010; Sue et al. 2018). Our study adds to this body of work by illustrating the manner in which a particular type of migration network—siblings—evolves, illustrating the need for the joint study of migration decisions within families to assess the extent to which they are linked and whether they are more consistent with theories assuming collective action. We hope that future studies and data collection efforts will continue to allow for such understanding of collective behaviors.

Notes

1. Sibling connections may also contribute to the diffusion of migrant networks across different sending areas if some siblings migrate internally before doing so internationally.

2. This research on the temporal and spatial expansion of sibling networks can also help to develop life-course perspectives on household models of migration decision-making, which had traditionally viewed sibling migrations primarily from the perspective of parental households (Stark and Bloom 1985; Massey et al. 1993). Recent research of Senegalese migration to Europe unearths new evidence that family migrant networks continue to influence individual migration, even once these individuals move out of their parental households (Liu and Lindstrom 2019).

3. We categorized California, Arizona, New Mexico, Texas, and Illinois as traditional destinations, and all other U.S. states as newer destinations.

4. In Mexico in 2013, men's average age at marriage was 30 and women's average age was 27 (INEGI 2018). In Senegal in 2010–11, women's average age at first marriage was 19.9, up from 16.8 in 1992–93 (United Nations 2014), while estimates for Senegalese men range between 25 and 30 years old (Pison et al. 1993; Garenne and Van de Walle 1989).

5. We should also note that the Mexican sample draws from more households in rural sending areas than the Senegalese survey, which only collected data in the Greater Dakar region and thus in mostly (but not exclusively) urban settings. This difference in composition and rural families may be more likely to engage in multi-sibling migration behaviors and thus the differences between MMP and MAFE.

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